# Study XYZ-001: Safety & Demographics Report

## Integrated Clinical Trial Summary — Example

Statistical Programming, Biostatistics

14 October 2026

## Introduction

This report summarises the **demographics** and **adverse event** profile for Study XYZ-001, a Phase II randomised controlled trial comparing **Drug A** versus **Placebo** in adult subjects with condition Z.

> **Note for programmers:** This document is a single `.qmd` source file. The tables, figures, and narrative are produced from the same R code every time the document is rendered — no copy-paste from SAS output.

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## Data

### Create Fictional Datasets

In a real study environment these datasets would be read from SAS transport files (`.xpt`) or from a validated ADaM data store on the clinical network. Here we create small fictional datasets inline so the example is **fully self-contained**.

```
# A tibble: 2 × 2
  TRT01P      n
  <chr>   <int>
1 Drug A     30
2 Placebo    30
```

```
[1] 75
```

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## Demographics Summary

The table below mirrors what you would produce in SAS with `PROC TABULATE` or `PROC REPORT`. Here it is built with `dplyr` and rendered with `kable()`.

Table 1: Table 1. Subject Demographics and Baseline Characteristics (Safety Population)

| Characteristic | Drug A (N=30) | Placebo (N=30) |
|---|---|---|
| Age (years), mean (SD) | 52.8 (13.8) | 48.8 (10.5) |
| Weight at baseline (kg), mean (SD) | 74.3 (11.8) | 77.8 (12.8) |
| Female, n (%) | 19 (63.3%) | 13 (43.3%) |
| White, n (%) | 18 (60.0%) | 23 (76.7%) |
| Black or African American, n (%) | 6 (20.0%) | 0 (0.0%) |
| Asian, n (%) | 5 (16.7%) | 5 (16.7%) |

# Adverse Events

## AE Incidence by Treatment Arm

Table 2: Table 2. Subjects with Any Adverse Event by System Organ Class and Preferred Term

| Adverse Event (Preferred Term) | Drug A (N=30) | Placebo (N=30) |
|---|---|---|
| Back pain | 3 (10%) | 2 (6.7%) |
| Dizziness | 8 (26.7%) | 2 (6.7%) |
| Fatigue | 5 (16.7%) | 3 (10%) |
| Headache | 4 (13.3%) | 4 (13.3%) |
| Insomnia | 5 (16.7%) | 4 (13.3%) |
| Nausea | 5 (16.7%) | 5 (16.7%) |
| Rash | 7 (23.3%) | 4 (13.3%) |
| Vomiting | 6 (20%) | 4 (13.3%) |

## AE Severity Profile

The figure below is produced inline by R — no external image file needed. In SAS you would export an ODS Graphics PNG and link it manually. Here the plot is part of the reproducible document.

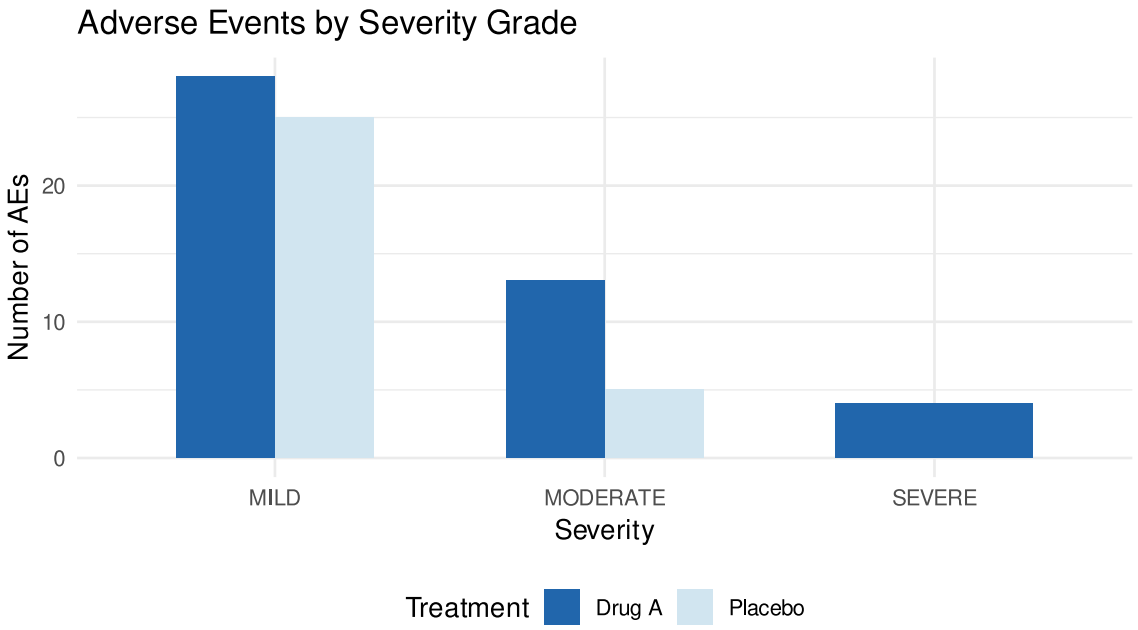



Figure 1: Figure 1. Adverse Event Counts by Severity and Treatment Arm

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## Inline R Values in Narrative Text

One of Quarto's most powerful features for clinical programming: values computed in R can be embedded directly in prose.

> A total of **60** subjects were included in the Safety Population: **30** in the Drug A arm and **30** in the Placebo arm. Across both arms, **45** subjects (75%) experienced at least one adverse event.

If the data change upstream and you re-render, **every number updates automatically**. No more manual edits after an unblinding data cut.

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## Key Takeaways for SAS Programmers

| SAS concept | Quarto / R equivalent |
|---|---|
| `ODS RTF` / `ODS PDF` | Change `format:` in YAML header |
| `%include 'setup.sas'` | `source("_setup.R")` or `library()` |
| `PROC REPORT` / `PROC TABULATE` | `dplyr` + `kable()` / `gt` / `flextable` |
| `ODS Graphics` | `ggplot2` — inline, no external file |
| Hard-coded counts in text | `60` — live inline values |
| Manual re-run after data cut | `quarto render` — full refresh in one command |

## Appendix: Session Info

Good practice: capture the R environment for reproducibility traceability — equivalent to logging SAS version and macro versions.

```
R version: R version 4.6.1 (2026-06-24)
```

```
Platform: x86_64-pc-linux-gnu
```

```
Key packages loaded:
```

```
  dplyr 1.2.1
  ggplot2 4.0.3
  knitr 1.52
  tidyr 1.3.2
```

```

Rendered: 2026-10-14 15:04:43 UTC
```